Lab 5: Static Code Analysis

Name:Pallavi Rajkumar Bubanale

SRN:PES1UG24CS314

Issues Identified and Fixes Applied

This table documents issues identified in the original movie seat reservation program and the changes made. Line numbers refer to the original code before fixes.

Issue	Type	Original line(s)	Description	Fix approach
Mutable default argument	Bug	9	The system_logs=[] default could cause the same list to be shared across function calls.	Changed the default to None and initialized a new list inside the function.
Missing docstrings and naming/style issues	Code Quality	Multiple	Functions lacked docstrings, and printSeatingSummary() did not follow snake_case naming conventions.	Added docstrings and renamed the function to print_seating_summary().
Broad exception handling	Code Quality / Security	21	A bare except could hide unexpected errors.	Replaced broad exception handling with a specific exception handler.
Weak MD5 hashing	Security	25	MD5 is unsuitable for security-sensitive hashing.	Replaced MD5 with SHA-256 for generating ticket identifiers.
Overly permissive file permissions	Security	55	Setting file permissions to 0o777 allowed overly broad access.	Removed the unnecessary os.chmod() call.
Unsafe file handling	Code Quality	28, 33	Files were opened without context managers or explicit encoding.	Used with open() and specified UTF-8 encoding.
Excess blank lines	Style	Multiple	Extra blank lines violated Flake8 formatting requirements.	Corrected blank lines between top-level functions.
Invalid seat numbers	Input Validation	Multiple	Negative seat numbers and nonnumeric values could be accepted as reservations.	Added validation to check seat numbers before reserving seats.

Final Tool Results

- Pylint: 10.00/10
- Bandit: No issues identified.
- Flake8: No errors reported.

Note: These results are recorded from the project documentation. SHA-256 is used here to create a short ticket identifier; it should not be treated as a secure authentication mechanism.